

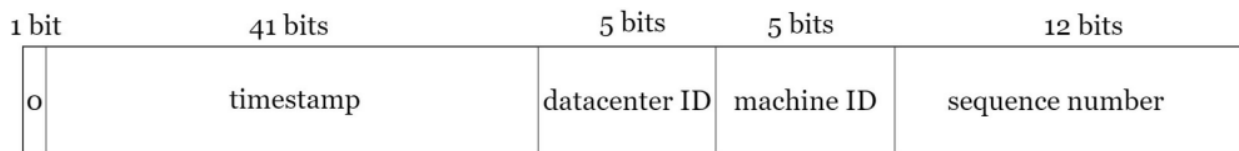
Design a Unique ID Generator - Twitter Snowflake Approach

1. What is Snowflake?

- A **64-bit unique ID generation system** developed by Twitter
- Designed for **distributed systems**

2. Core Idea

- Use **divide and conquer**
- Split a 64-bit ID into multiple parts



3. 64-bit ID Structure

Components

Sign Bit (1 bit)

- Always **0**
- Reserved for future use

Timestamp (41 bits)

- Time in **milliseconds since custom epoch**
- Default epoch:

None

1288834974657 (Nov 04, 2010, 01:42:54 UTC)

- Ensures **time ordering**

Datacenter ID (5 bits)

- Supports:

None

$2^5 = 32$ datacenters

Machine ID (5 bits)

- Supports:

None

32 machines per datacenter

Sequence Number (12 bits)

- Supports:

None

$2^{12} = 4096$ IDs per millisecond per machine

- Resets every millisecond

4. How It Works

- Datacenter ID + Machine ID → fixed at startup
- Timestamp → generated at runtime
- Sequence number → increments per request

5. Pros

- Generates **64-bit numeric IDs**
- IDs are **time-ordered**
- Works in **distributed systems**
- High throughput

6. Cons

- Requires careful configuration of machine & datacenter IDs
- Clock synchronization issues can cause problems

7. Key Takeaway

Snowflake:

- Meets all requirements:

- Unique
- 64-bit
- Time-ordered
- Distributed